

## Homework Assignments for MAT 1001, Fall, 2025

- 2.1: #1, 7, 8, 15, 16  
2.2: #1, 4, 9, 10, 18, 19, 23, 28, 32, 40, 44, 45, 54, 57, 65 (a), 75, 77, 80  
2.3: skipped  
2.4: #1, 5, 6, 17, 20, 28, 32, 34, 44, 45, 46  
2.5: #7-10, 26, 27, 32, 38, 43, 47, 55, 65, 67, 70  
2.6: #2, 10, 11, 16, 21, 23, 31, 36, 39, 43, 63, 67, 69, 70, 80, 82, 99
- 3.1: #1, 2, 23, 33, 41, 47  
3.2: #1, 20, 27-31, 34, 43, 44, 46-48, 57, 58  
3.3: #6, 11, 17, 20, 21, 29, 37, 38, 41, 43, 48, 55-58  
3.4: #3, 7, 15, 21, 23, 25, 26  
3.5: #3, 12, 15, 17, 26, 32, 34, 45, 50, 51, 56, 58, 59  
3.6: #5, 7, 10, 15, 29, 34, 41, 54, 65, 70, 73, 74, 82, 84,  
3.7: #1, 12, 19, 29, 41  
3.8: #7, 23, 27, 32, 42  
3.9: #1, 6, 17, 18, 29, 37, 42, 51, 56
- 4.1: #1-6, 20, 36, 42, 45, 50, 51, 67, 68  
4.2: #13, 15 a, (iii), 15b, 18, 20, 23, 27, 33, 60, 61, 63, 65  
4.3: #15-18, 23, 35, 55a, 65  
4.4: #1, 4, 17, 23, 32, 46, 73, 76, 93, 95, 98, 99, 104  
4.5: #1, 7, 12, 20a, 22, 28, 38, 48, 49, 51, 54, 67a  
4.6: #1, 3, 7, 9, 10, 20  
4.7: #1, 7, 12, 23, 32, 34, 52, 59, 63, 75, 85, 95, 99, 103
- 5.1: #1, 3, 5, 7, 9, 19  
5.2: #8, 12, 30, 32, 33, 35, 37, 38, 39, 40  
5.3: #1, 3, 6, 8, 9, 17, 20, 55, 71, 73, 76, 77, 78, 81, 83  
5.4: #1, 3, 4, 8, 9, 10, 12, 17, 22, 24, 27, 28, 30, 40, 44, 46, 61, 63, 68, 71, 73  
5.5: #18, 20, 21, 22, 26, 31, 38, 42, 43, 46, 56, 62  
5.6: #8, 11, 12, 21, 23, 25, 59, 63, 71, 72, 78, 82, 83, 84-87
- 6.1: #1, 10, 12, 14, 19, 23, 27, 29, 37, 41, 43, 51, 59  
6.3: #1, 2, 9, 23, 30  
6.4: #13, 18, 22, 23, 25, 28  
6.5: #2, 7, 10, 14, 23, 24, 29, 36, 44  
(6.2 and 6.6 skipped)
- 7.1: #1-6, 12, 14, 17, 25, 31, 38, 39, 44, 56, 59  
7.2: #1, 4, 5, 8, 9, 21, 23, 34, 39, 42, 45, 53, 56, 65, 70, 73, 77, 85

7.3: #1, 8, 14, 16, 24, 26, 27, 29, 38, 48, 56, 59, 62, 65, 74, 75, 82, 85, 88,  
91, 96, 99, 111, 118, 121, 123, 131, 136

7.4: delayed

7.5: #16, 20, 21, 26, 31, 38, 40, 41, 45, 51, 53, 59, 60, 61, 66, 67, 68, 75, 85, 88

7.8: (taught before 7.6) #1, 9-12, 20, 21(a), 22

7.6: #1, 3, 5, 7, 13, 16, 18, 22, 25, 33, 44, 47, 52, 63, 65, 69, 74, 87, 91, 92, 96, 97, 123,  
127

8.1: skipped

8.2: #2, 6, 12, 17, 23, 28, 29, 31, 32, 37, 43, 45, 50, 51, 53, 61, 63, 69

8.3: #3, 14, 15, 19, 20, 24, 27, 30, 33, 34, 35, 53, 54, 56, 64, 67

8.4: #2, 3, 5, 8, 9, 10, 15, 17, 21, 26, 36, 37, 57, 58

8.5: #6, 7, 13, 16, 20, 23, 27, 31, 34, 35, 41

8.7: #8, 24, 25, 28, 29, 31

8.8: #4, 10, 14, 20, 23, 27, 32, 35, 39, 41, 52, 53, 56, 61, 64, 65, 67, 72, 76

9.1: #1-6, 11, 14 (do 11 and 14 after learning 7.4 and 9.2)

7.4: #11, 12, 13, 17, 30, 34, 42

9.2: #1, 4, 16, 17

9.3: #13, 15

9.4: #1, 7, 10, 12, 14, 18 (for these problems, no need to study the sign of the second derivative of the solutions)